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TOPIC 1: INTRODUCTION TO AUTONOMOUS VEHICLES & LEVELS OF AUTONOMY

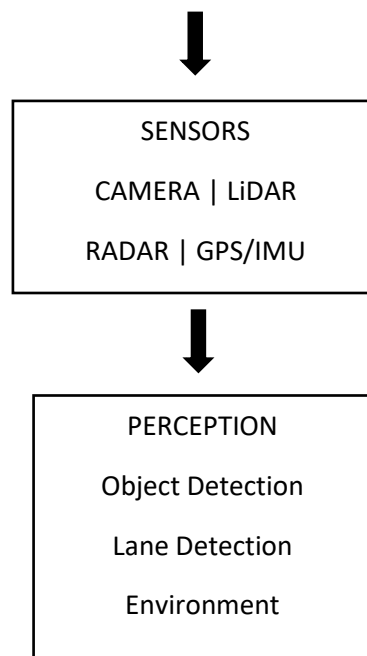
PROJECT: Functional Architecture of a Level 4 Autonomous Vehicle

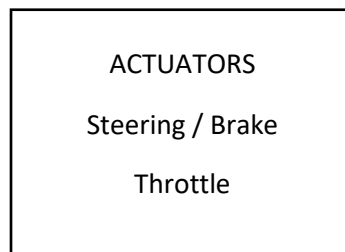
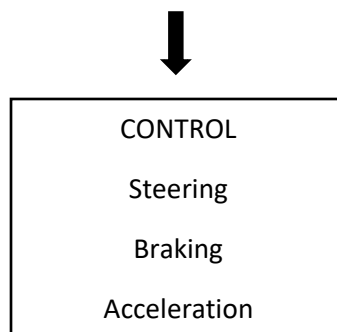
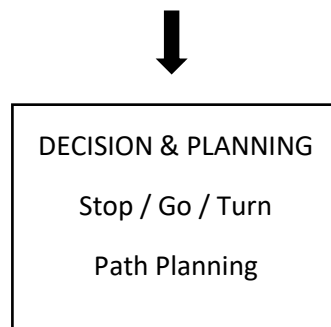
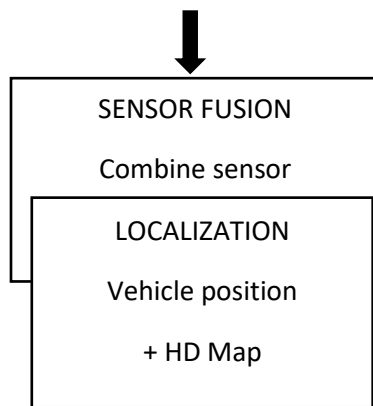
1. OBJECTIVE:

The objective is to design a functional architecture showing how a Level 4 autonomous vehicle senses its surroundings, makes driving decisions, plans a safe path and controls the vehicle without human intervention within its defined operating conditions.

2. FUNCTIONAL ARCHITECTURE:

LEVEL 4 AUTONOMOUS VEHICLE





3. MODULE DESCRIPTION:

SERIAL NUMBER	<u>MODULE</u>	<u>FUNCTION</u>
1.	Sensors	Collect information about the road and surroundings.
2.	Perception	Detects vehicles, pedestrians, lanes and obstacles.
3.	Sensor Fusion	Combines information from multiple sensors.
4.	Localization	Determines the vehicle's position.
5.	Decision and Planning	Decides actions and selects a safe driving path.
6.	Control	Converts the planned path into steering, braking and acceleration commands.
7.	Actuators	Physically control steering, brakes and acceleration.

4. CONCLUSION:

The proposed architecture allows a Level 4 autonomous vehicle to sense its environment, understand the situation, make decisions and control the vehicle. The system can perform these driving functions without continuous human control within its defined operational conditions.

SIMULATION OVERVIEW

A simple Python-based simulation was developed to demonstrate the working of a Level 4 autonomous vehicle in a basic driving scenario. The simulation focuses on a pedestrian detection and automatic braking situation.

In the simulated scenario, the autonomous vehicle initially moves forward at a constant speed. After a specified time, a pedestrian detection event is simulated. The vehicle then changes its decision from CONTINUE to STOP and applies automatic braking.

The simulation demonstrates how the decision-making and control modules of an autonomous vehicle can respond to a detected obstacle. It provides a simplified representation of the interaction between perception, decision-making, and vehicle control.

SIMULATION OBJECTIVE

The main objective of the simulation is to demonstrate the basic decision-making and control behaviour of a Level 4 autonomous vehicle in a pedestrian detection scenario.

The specific objectives are:

- To simulate the movement of an autonomous vehicle over time.
- To represent a pedestrian detection event during vehicle movement.
- To demonstrate the change in vehicle decision from CONTINUE to STOP.
- To simulate automatic braking after pedestrian detection.
- To observe the reduction in vehicle speed during braking.
- To calculate and visualize the distance travelled by the vehicle.
- To demonstrate the interaction between perception, decision-making, and control functions.

SIMULATION SCENARIO

The simulation represents an autonomous vehicle travelling on a road while monitoring its surroundings. The vehicle starts with an initial speed of 20 m/s.

During the initial part of the simulation, no immediate obstacle is considered to be present. Therefore, the vehicle continues moving and the autonomous driving decision remains CONTINUE.

At 4 seconds, a pedestrian detection event is simulated. After this event, the decision-making module changes the vehicle action from CONTINUE to STOP.

The control module then simulates automatic braking by gradually reducing the vehicle speed. The vehicle eventually reaches a speed of 0 m/s, representing a complete stop.

The overall scenario can be represented as such:

Vehicle Moving -> Pedestrian Detected -> STOP Decision -> Automatic Braking -> Vehicle Stops

The pedestrian detection in this simulation is a predefined event used to demonstrate the autonomous vehicle's decision and control response. It does not represent actual pedestrian detection using a physical camera, LiDAR, or Radar sensor.

SIMULATION RESULT

The simulation successfully demonstrates the basic autonomous vehicle response. Initially, the vehicle continues moving at 20 m/s. When the pedestrian detection event occurs at 4 seconds, the system changes the decision to STOP and gradually reduces the vehicle speed to 0 m/s.

This confirms that the decision-making and braking logic work as expected in the simulated scenario.

CONCLUSION

This simulation demonstrates a basic autonomous driving scenario in which the vehicle detects a predefined pedestrian event, changes its decision from CONTINUE to STOP, and applies automatic braking until the vehicle comes to a complete stop.